Practical No :03

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Problem Statement : Write a program to solve a fractional Knapsack problem using a greedy method.

Code :

```
def fractional_knapsack(capacity, items):
    # Sort items by value-to-weight ratio in descending order
    items.sort(key=lambda item: item[1] / item[0], reverse=True)

    total_value = 0

    for weight, value in items:
        if capacity == 0:
            break

        if weight <= capacity:
            # Take the entire item
            capacity -= weight
            total_value += value
        else:
            # Take the fraction of the item that fits
            total_value += value * (capacity / weight)
            capacity = 0

    return total_value


n = int(input("Enter number of items: "))

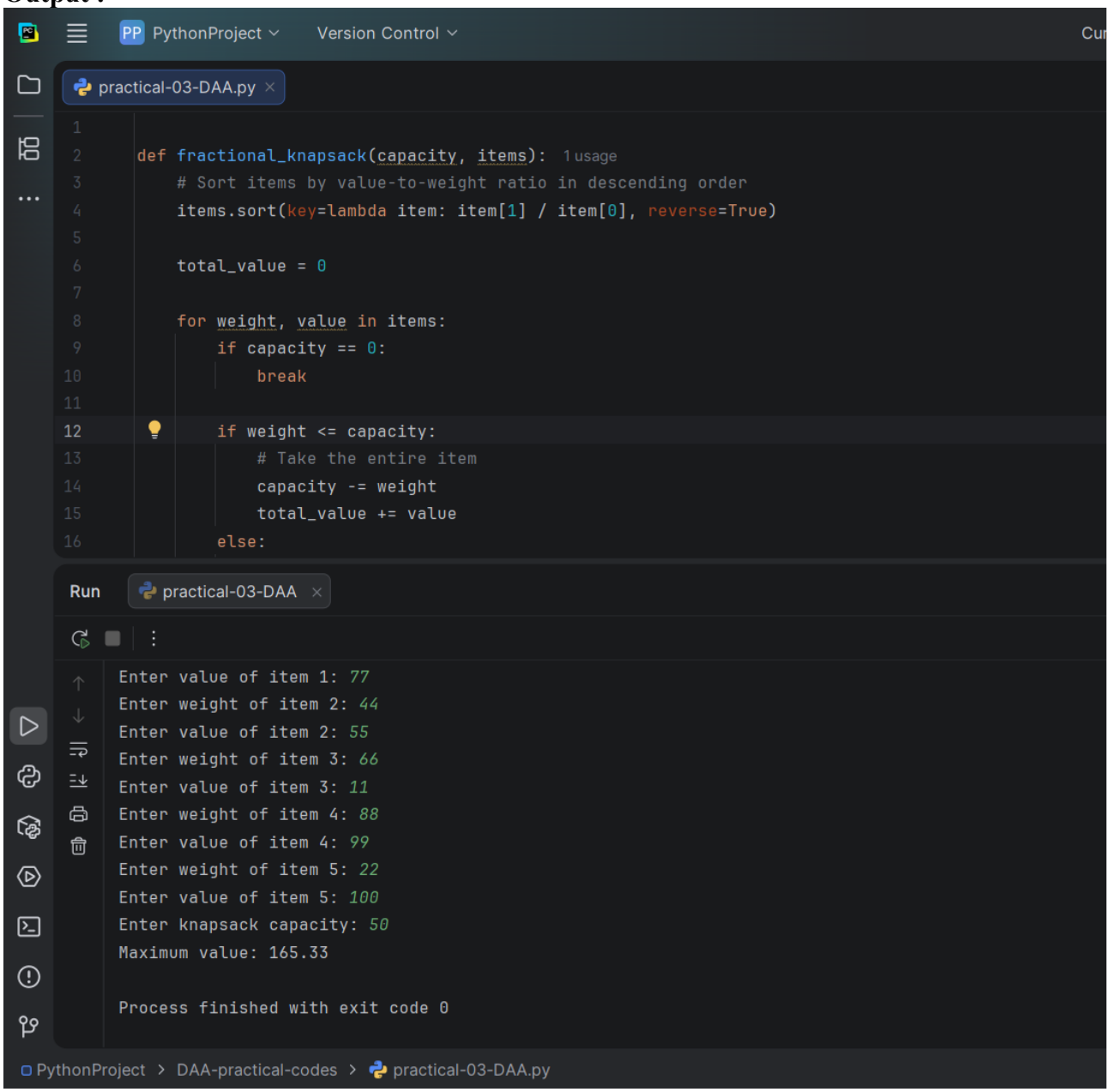
items = []
for i in range(n):
    weight = float(input(f"Enter weight of item {i + 1}: "))
    value = float(input(f"Enter value of item {i + 1}: "))
    items.append((weight, value))

capacity = float(input("Enter knapsack capacity: "))

max_value = fractional_knapsack(capacity, items)

print("Maximum value:", round(max_value, 2))
```

Output :


PythonProject
Version Control
practical-03-DAA.py
def fractional_knapsack(capacity, items): 1 usage
Sort items by value-to-weight ratio in descending order
items.sort(key=lambda item: item[1] / item[0], reverse=True)
total_value = 0
for weight, value in items:
if capacity == 0:
break
if weight <= capacity:
Take the entire item
capacity -= weight
total_value += value
else:
Run
practical-03-DAA
Enter value of item 1: 77
Enter weight of item 2: 44
Enter value of item 2: 55
Enter weight of item 3: 66
Enter value of item 3: 11
Enter weight of item 4: 88
Enter value of item 4: 99
Enter weight of item 5: 22
Enter value of item 5: 100
Enter knapsack capacity: 50
Maximum value: 165.33
Process finished with exit code 0
PythonProject
DAA-practical-codes
practical-03-DAA.py